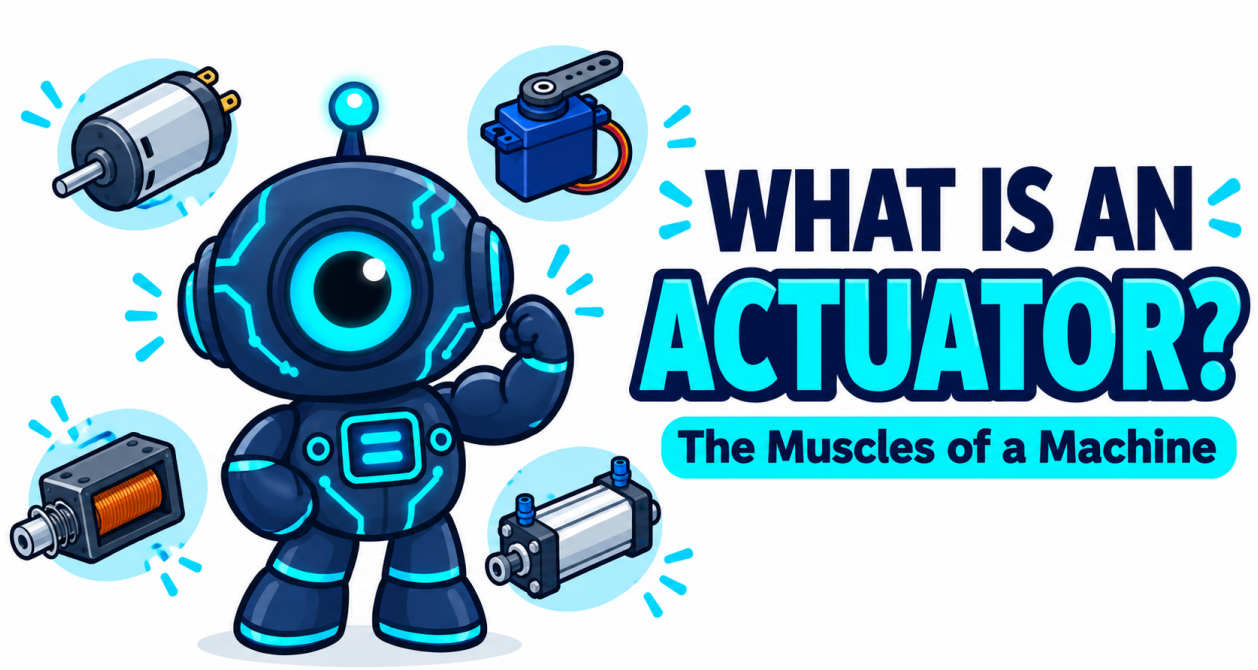
AI MECH LAB

Mechatronics Course — Beginner Stage

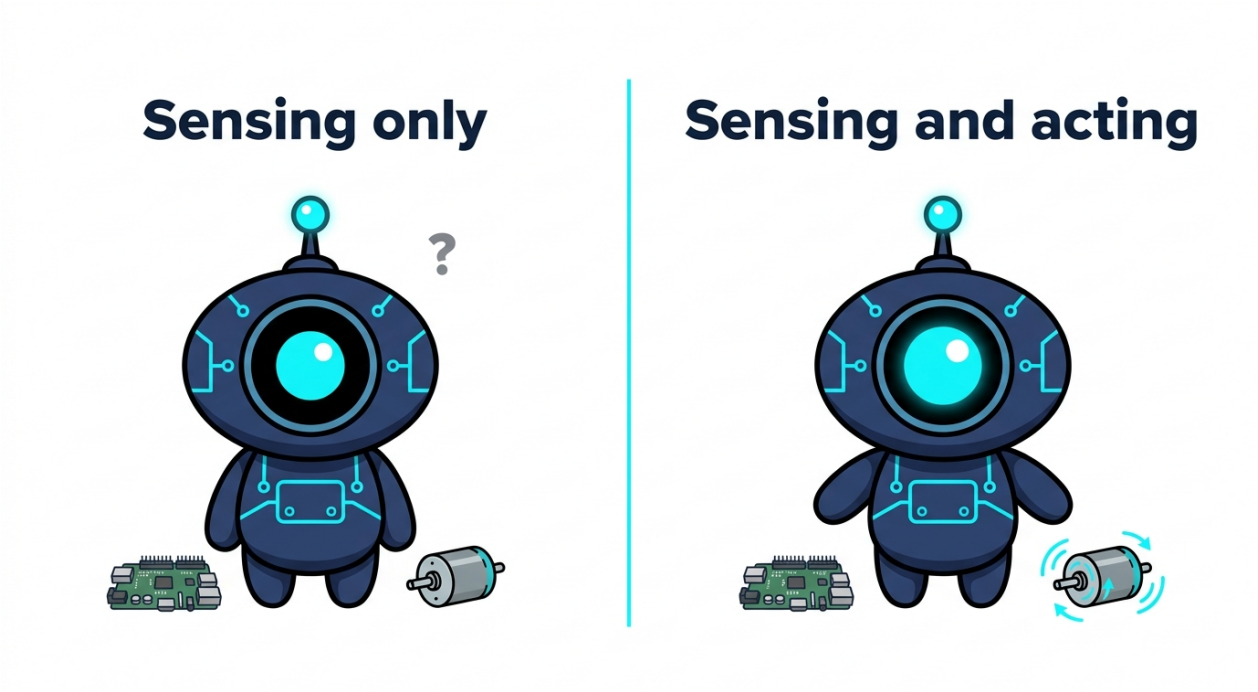


Module 4 · Topic 4.1

What Is an Actuator?

What Is an Actuator?

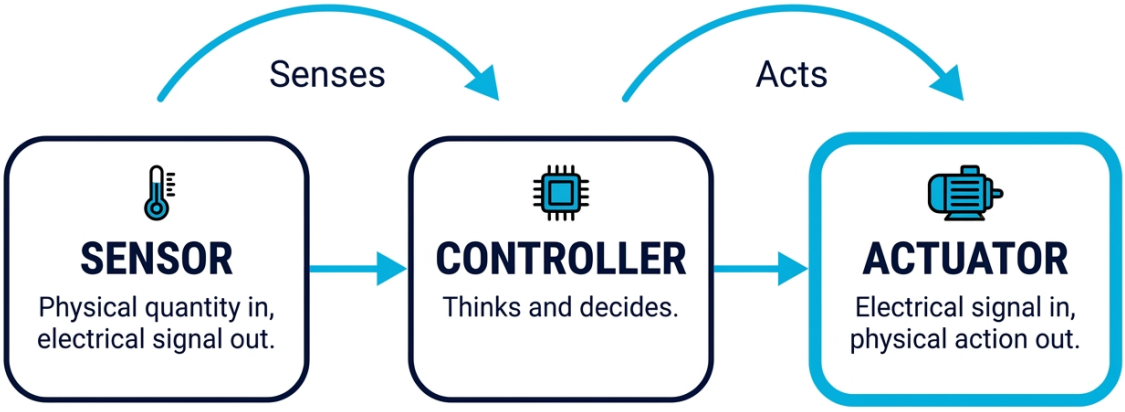
Module 3 gave your controller senses. It can now measure temperature, distance, light and force. But a controller that only senses is an expensive thermometer. To change anything in the world, it has to act, and the devices that do the acting are actuators.



A controller that only senses cannot change anything. Add an actuator and it can act.

What an actuator is

An actuator converts energy, most often electrical, into physical action: motion, force, heat, light or sound. It is the mirror image of a sensor. A sensor turns a physical quantity into an electrical signal. An actuator turns an electrical signal into a physical effect. Between them sits the controller, and together they form the sense-think-act loop from Topic 3.1. This module is about the last link.

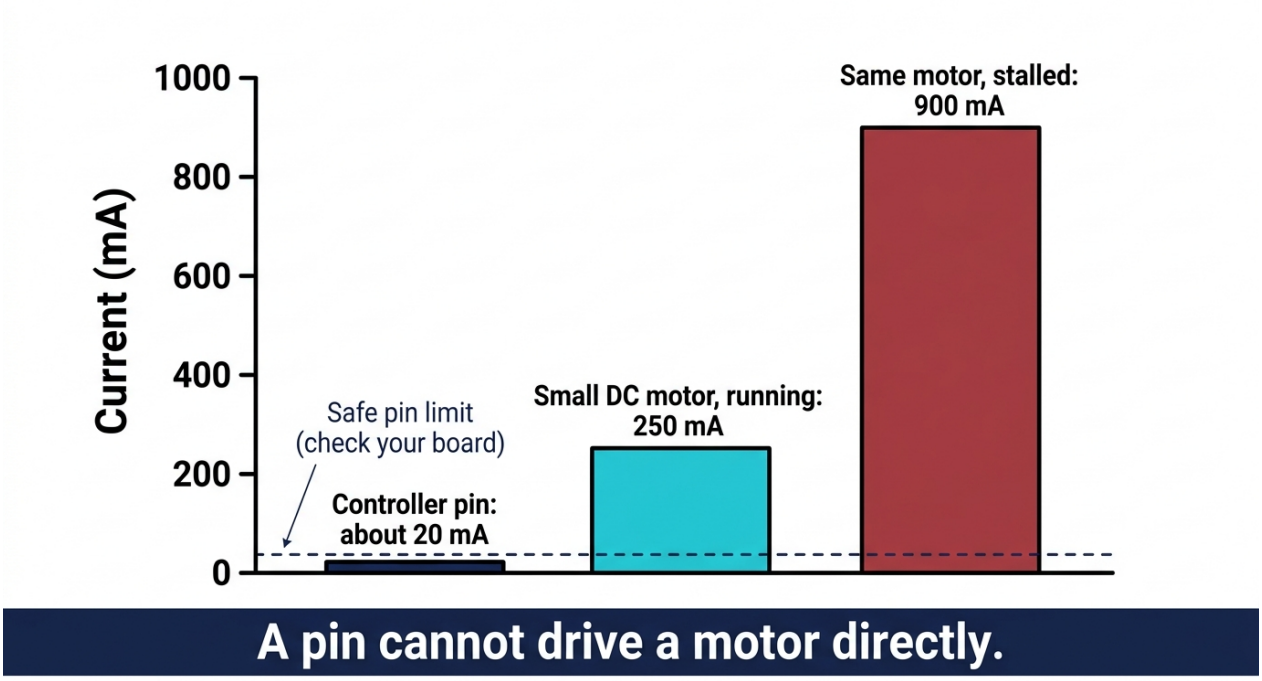


Sensor, controller, actuator: the sense-think-act loop. This module focuses on the actuator.

The word covers a wide range of devices. A motor spinning a wheel is an actuator. So is a solenoid that pops open a door lock, a heating element in an oven, a pump pushing water through a pipe, and a pneumatic cylinder clamping a part on a factory line.

Why a controller pin cannot drive one

Here is the first problem every beginner hits. A microcontroller pin is a signal source, not a power source. On an Uno-class board, a pin can supply around 20 mA at 5 V, and your own board's datasheet gives the exact limit. Compare that with the motor from Topic 2.4, which drew 250 mA while running and 900 mA when stalled. Even a small motor wants many times what a pin can give.



A controller pin supplies tens of milliamps. A small motor needs hundreds. The figures are illustrative.

Connect the two directly and one of them fails, and it is not the motor. The pin overheats and the controller can be damaged or destroyed. So a controller never powers an actuator itself. It controls something that does.

The drive chain

The solution is a three-part chain, with a separate power supply.

1. The **controller** sends a small control signal, a few milliamps at most.
2. A **driver** receives that signal and switches or regulates a much larger current. The driver can be a single transistor from Module 2, a relay, or a dedicated motor-driver chip.
3. The **actuator** receives that larger current from a power supply that is separate from the controller's pins.

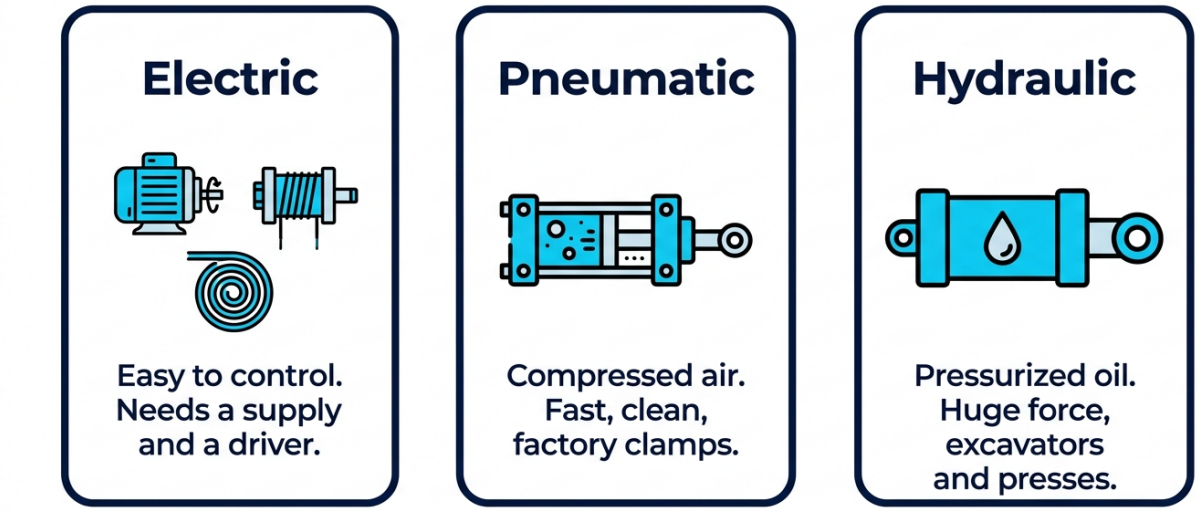
The controller and the driver's supply must share a common ground, the same rule you met in Topic 3.3, or the control signal has no reference and the driver behaves unpredictably. Topic 4.2 shows how to build this chain for a DC motor.

Where actuators get their energy

Electric actuators are the ones you will use in this course: motors, solenoids, heating elements and buzzers. They are easy to control from a controller and need only a power supply and a driver.

Pneumatic actuators run on compressed air. A common form is a cylinder with a piston that extends when air is let in through a valve. They are fast, simple and clean, which is why they are common in factory clamps and pick-and-place machines. Air is compressible, so they are poor at holding a position precisely in the middle of their stroke.

Hydraulic actuators work the same way but use pressurized liquid, usually oil. Liquid hardly compresses, so these produce very large forces and hold their position firmly. Excavator arms and presses use them. The cost is a pump, hoses and the risk of leaks.



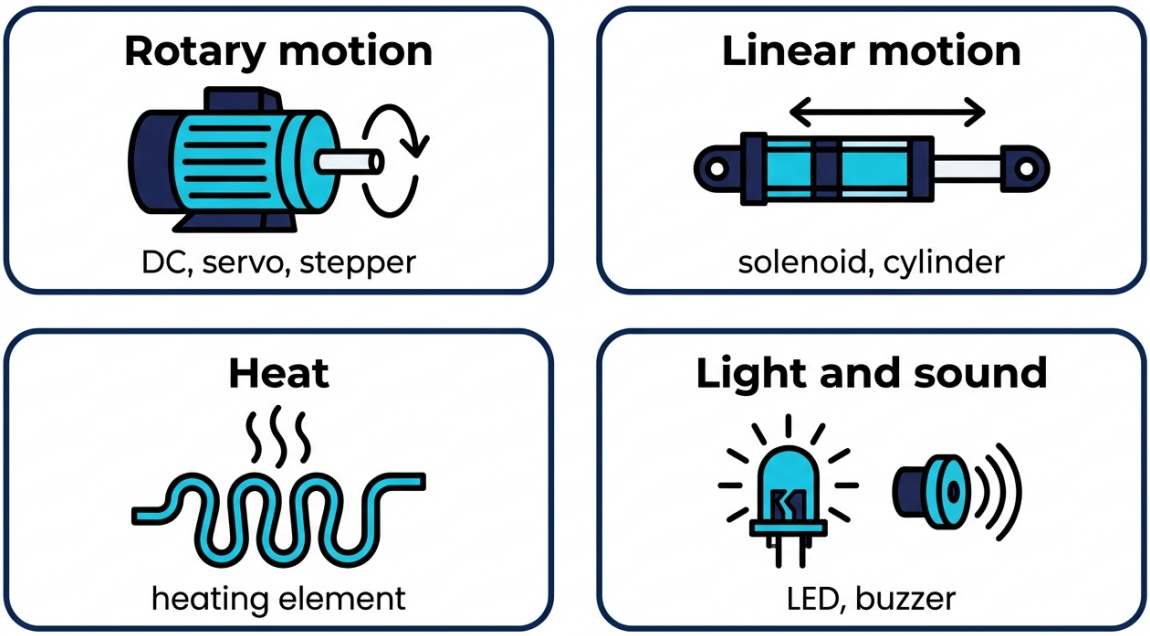
Three energy sources: electric, pneumatic and hydraulic.

A controller drives both pneumatic and hydraulic actuators the same way: it switches a valve, usually through an electric driver.

What actuators output

Actuators sort by what they produce.

- **Rotary motion:** DC motors, servo motors and stepper motors.
- **Linear motion:** solenoids, linear actuators and pneumatic or hydraulic cylinders.
- **Heat:** heating elements, as in ovens and 3D printer hotends.
- **Light and sound:** LEDs and buzzers, usually called output devices, though they follow the same chain.



Actuators sorted by what they produce.

Two properties matter for anything that moves. **Force** is the push or pull, measured in newtons. **Torque** is the equivalent for rotation, a twisting force measured in newton-metres. A motor that can spin fast but cannot turn a heavy load is no use, so you always ask both how much and how fast. Gearing trades one for the other: a gearbox lowers speed and raises torque. Topic 4.2 returns to this.

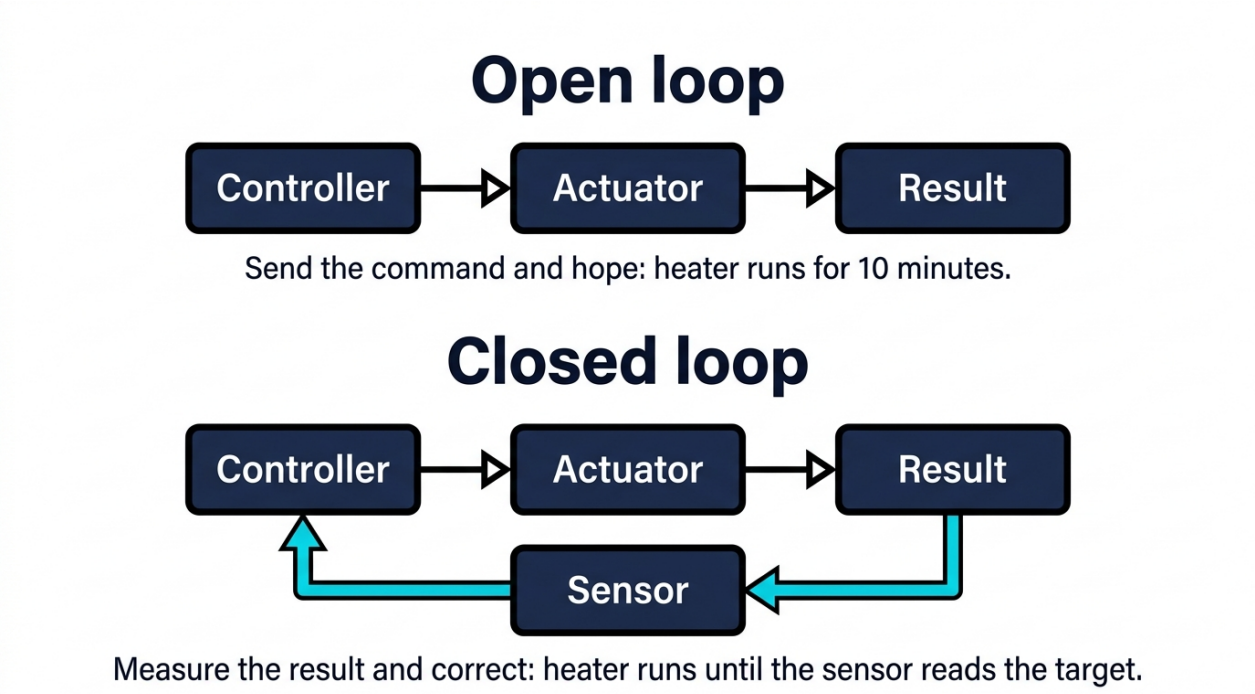
Controlling an actuator

There are three ways to control one.

On and off. A transistor or relay either passes current or does not. This is enough for a heater, a solenoid or a fan that only needs to run or stop.

Variable. To change a motor's speed, you vary the average power it receives. Controllers do this with PWM, which switches the supply on and off thousands of times per second. The fraction of time it is on, the duty cycle, sets the average power. At 50% duty, a motor behaves roughly as though it were running at half the voltage. Topic 4.2 covers PWM properly.

With feedback. Open-loop control sends a command and hopes it worked. Closed-loop control adds a sensor that measures the result and feeds it back, so the controller can correct itself. A heater that runs for ten minutes regardless of temperature is open-loop. A heater that runs until a temperature sensor reads the target is closed-loop. Closed loops are more accurate and more robust to changing conditions, and they are why Modules 3 and 4 belong together.



Open loop sends a command and hopes. Closed loop measures the result and corrects.

Two hazards to know now

Stall current. A motor that is jammed or overloaded stops turning and draws far more current than it does when running, which is the 900 mA case from Topic 2.4. Choose your driver for the stall current, not the running current.

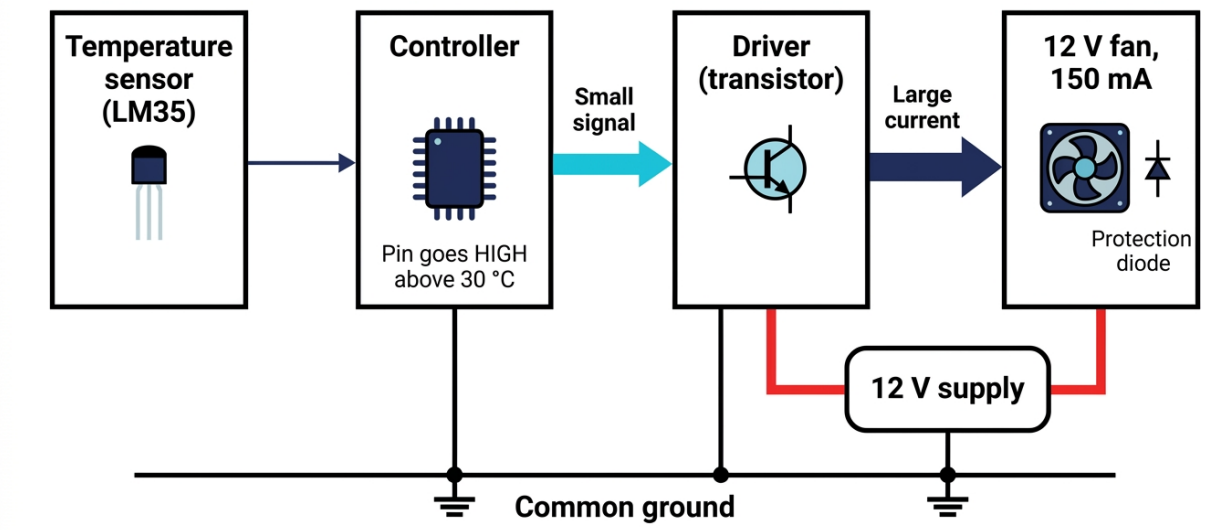
Inductive kick. Motors, solenoids and relay coils store energy in a magnetic field. When you cut their current suddenly, they push back with a voltage spike that can be several times the supply and can destroy a transistor. A diode connected across the coil gives that energy a safe path. This is a job for the diodes from Topic 2.3, and Topic 4.2 shows where to place one.

Keep your fingers clear of anything that moves, switch power off before changing wiring, and treat heating elements as hot even after they are off. Never use a relay to switch mains electricity unless you are trained and the parts are rated for it.

A worked example: a temperature-controlled fan

Say you want a small cooling fan to switch on when a box gets too warm. The fan is a 12 V DC type that draws 150 mA. These figures are illustrative, not from a specific part.

A controller pin gives about 20 mA at 5 V. The fan needs 7.5 times that current at a different voltage, so the pin cannot drive it directly. The chain looks like this. An LM35 reads the temperature. The controller compares it with a threshold of 30 °C. When it is exceeded, the controller sets a pin high. That pin feeds a transistor, which switches the fan on from a separate 12 V supply. A protection diode sits across the fan, and the controller and supply share ground. When the temperature drops, the pin goes low and the fan stops.

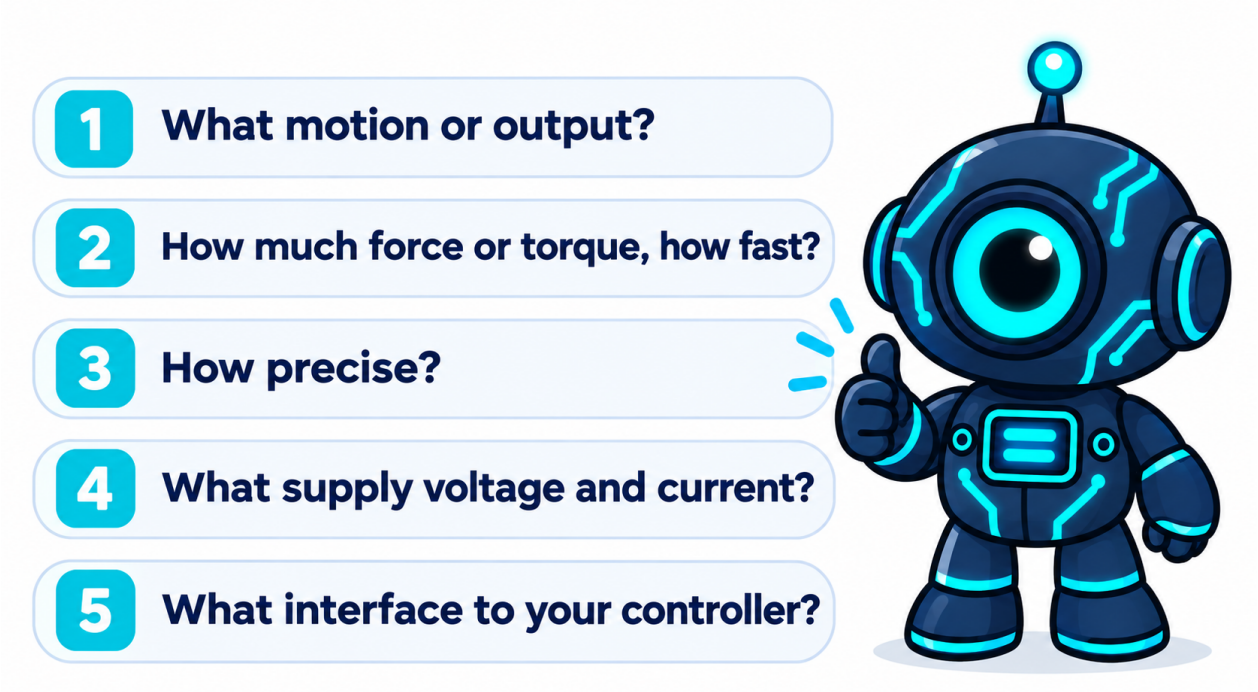


The fan example as a block diagram. The exact transistor and diode wiring is built in Topic 4.2.

One refinement. If the fan turns on at 30 °C and off at 30 °C, the noise from Topic 3.3 makes it flicker as the reading hovers around the threshold. The standard fix is to turn on at 30 °C and off at 28 °C. That gap is called hysteresis, and it is why a closed loop works smoothly instead of chattering.

Choosing an actuator

Work through five questions. What motion or output do you need? How much force or torque, and at what speed? How precise must it be? What supply voltage and current does it need? What interface does it need from your controller?



Five questions to ask before choosing an actuator.

Take a gate that must push boxes off a conveyor. The push is short, fast and on or off, so a solenoid or a small pneumatic cylinder fits. If instead you need an arm that swings to a chosen angle, a servo fits. For a belt that must run continuously and know its position, a DC motor with an encoder fits.

Quick reference

Actuator	Output	Typical control	Typical use
DC motor	Continuous rotation	Voltage or PWM through a driver	Fans, wheels, pumps
Servo motor	Rotation to a commanded angle	Pulse signal, internal feedback	Hobby arms, steering
Stepper motor	Rotation in fixed steps	Step signals through a driver	3D printers, positioning
Solenoid	Short push or pull	On/off through a driver	Locks, valves, gates
Relay	Switches a separate circuit	On/off through a driver	Heaters, heavy loads
Heating element	Heat	On/off or PWM through a driver	Ovens, 3D printer hotends
Pneumatic cylinder	Linear force from air	Valve, on/off	Factory clamps, pick-and-place
Hydraulic cylinder	Very large linear force	Valve control	Excavators, presses

> ***Going deeper:*** *the Robotics course looks at actuators for machines that move through the world, comparing the options for wheels and joints and how they work together.*

Where this leaves you

You now know what an actuator is, why a controller needs a driver to power one, and how to sort actuators by energy source and output. You can choose between on/off, variable and closed-loop control, and you know the two hazards that destroy parts: stall current and inductive kick. Next comes Topic 4.2, where we build the drive chain for the most common actuator, the DC motor, including the transistor, the protection diode and PWM speed control.